

Boundary Value Analysis

A technique focused on boundaries between partitions, where defects are often found.

Idea

Test values exactly at the boundary and immediately below/above when applicable.

Example

For the range 18–65, consider 17, 18, 19, 64, 65 and 66 according to the rule and risk.

Applications

Numeric fields, dates, sizes, quantities, pagination, upload limits and thresholds.

Combination

Use with equivalence partitioning and business rules to avoid a mechanical list of numbers.

When to use

Use this knowledge when risk, architecture or the development lifecycle stage requires specific validation. Before applying it, confirm the objective, environment, required data and expected evidence.

How QA applies it in practice

Start from the behavior or risk that needs validation, prepare data and environment, execute the technique or tool reproducibly and record evidence that allows the team to understand the result.